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PY211 Lab 3

Objective: Determine the spring constant K (N/m) of a given spring.

Equipment:

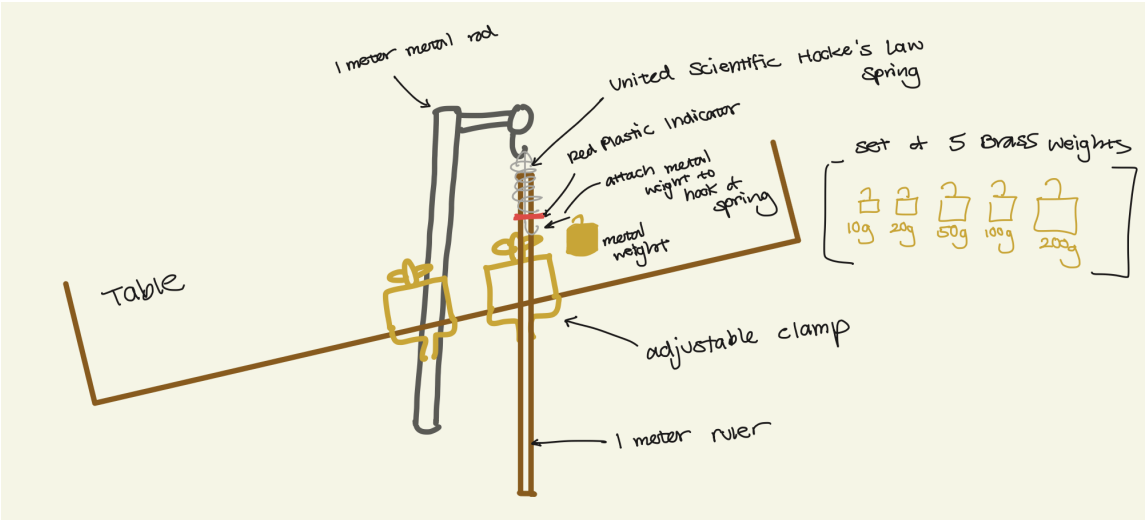
- 5 brass weighs of varying masses (10g, 20g, 50g, 100g, 200g)
- Meter stick
- 2 clamps
- 1 Metal Rod
- United Scientific Hooke's Law Metal Spring
- Red Plastic indicator

Setup:

- Secure the two clamps to the same horizontal height surface (like a table)
- Insert the Meter stick into one clamp so that it is perpendicular to the table
- Insert the metal rod in the other clamp so that it is perpendicular to the table, parallel to the meter stick, and so that its maximum height is slightly taller than the meter stick's
- Hang the spring from one end to the top of the metal rod
- Attach the red plastic indicator to the lower end of the spring
- Adjust the clamps of the metal rod as necessary until the top of the Meter stick (x=0) is aligned with the red indicator

Process:

1. Attach the lowest mass weight to the metal hook attached to the lower end of the spring
2. Record the displacement indicated by the red indicator on the meter stick
3. Remove the weight and repeat steps 1-2 with remaining weights of increasing masses until all have been tested



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Assumptions

- The gravitational acceleration constant in this location is 9.80665
- The reported masses of the brass weights are correct
- The alignment of the meter stick to the red indicator to assert x=0 was correct and had no error
- The spring during the trials maintained its elastic properties and was not stretched beyond its elastic region
- The meter stick and the metal rod were perfectly perpendicular to the table and parallel to each other
- The spring operates ideally in the linear regime

Sources of Experimental Uncertainty & Steps to Minimize

- Possible misreading of the spring's displacement → we made the best effort to align the red indicator as close to the tick marks of the meter stick as possible to get the most accurate read
- Wobbling or shaking of the table or the clamp fixture which may have caused an inconsistent position reading → we measured the displacement when the apparatus was as stable as possible
- Determining the precise initial position → positioned the red plastic indicator as close to the top of the meter stick as possible to set the initial position accurately

Results of Experiment

Mass (kg)	Extension (m)	Delta Extension (m)
0.01	0.003	0.001
0.02	0.007	0.001
0.05	0.019	0.001
0.1	0.04	0.001
0.2	0.08	0.001

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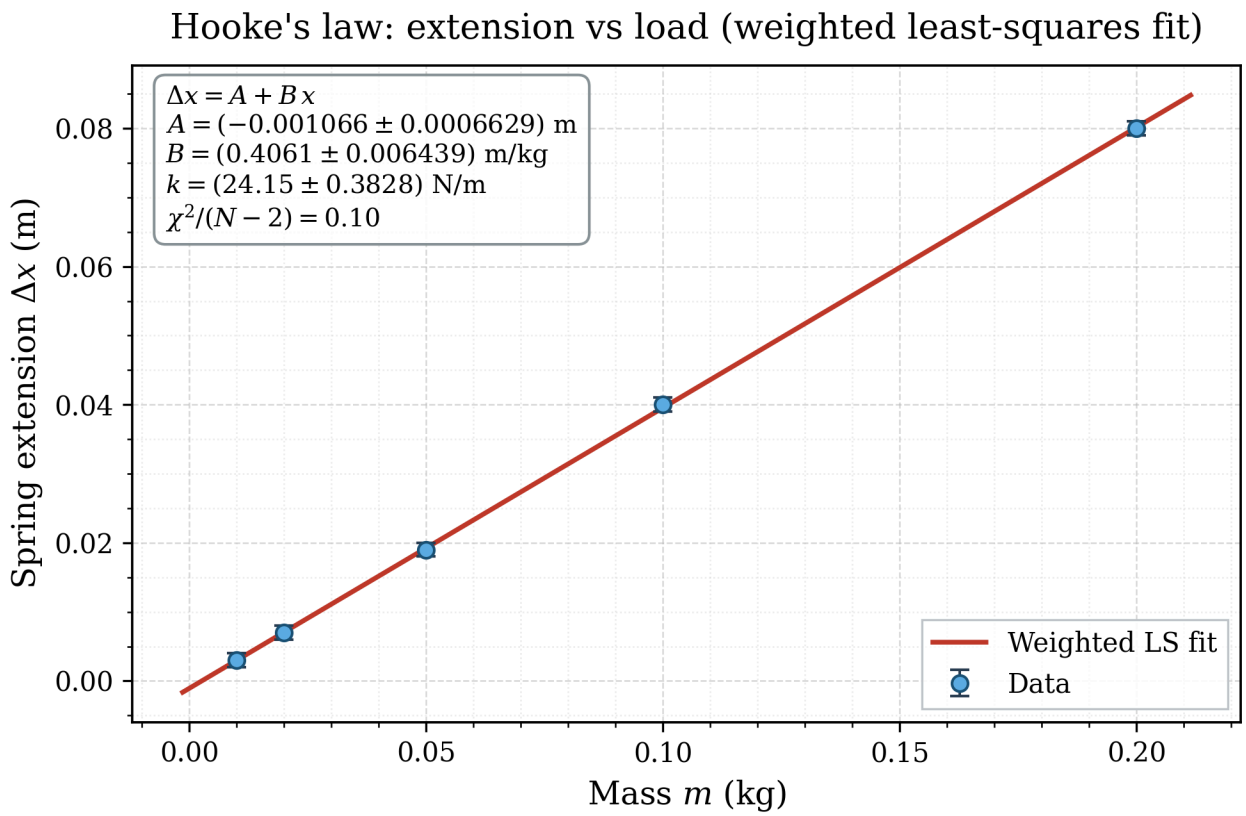
Graphical Analysis



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Calculations

Physical model

$$F = k\,\Delta x$$
$$mg = k\,\Delta x \quad \Rightarrow \quad \Delta x = \frac{g}{k}\,m$$

So the graph of Δx vs. m should be linear with slope $B = g/k$, hence

$$k = \frac{g}{B}$$

Weighted least-squares coefficients

Intercept and slope:

$$A = \frac{\left(\sum_i w_i x_i^2\right)\left(\sum_i w_i y_i\right) - \left(\sum_i w_i x_i\right)\left(\sum_i w_i x_i y_i\right)}{\left(\sum_i w_i\right)\left(\sum_i w_i x_i^2\right) - \left(\sum_i w_i x_i\right)^2} \tag{1}$$

$$B = \frac{\left(\sum_i w_i\right)\left(\sum_i w_i x_i y_i\right) - \left(\sum_i w_i x_i\right)\left(\sum_i w_i y_i\right)}{\left(\sum_i w_i\right)\left(\sum_i w_i x_i^2\right) - \left(\sum_i w_i x_i\right)^2} \tag{2}$$

Uncertainties on A and B

$$\delta_A = \sqrt{\frac{\sum_i w_i x_i^2}{\left(\sum_i w_i\right)\left(\sum_i w_i x_i^2\right) - \left(\sum_i w_i x_i\right)^2}} \tag{3}$$

$$\delta_B = \sqrt{\frac{\sum_i w_i}{\left(\sum_i w_i\right)\left(\sum_i w_i x_i^2\right) - \left(\sum_i w_i x_i\right)^2}} \tag{4}$$

From slope to k and δk (error propagation)

$$k = \frac{g}{B} \qquad \Rightarrow \qquad \delta k = \left|\frac{\mathrm{d}k}{\mathrm{d}B}\right| \delta_B = \frac{g}{B^2}\,\delta_B$$

Weighted chi-squared:

$$\chi^2 = \sum_i w_i \left(y_i - A - B x_i\right)^2 \tag{5}$$

Numerical results (weighted least-squares fit)

Fit model: spring extension vs. mass,

$$\Delta x = A + Bm, \qquad m \text{ in kg}, \quad \Delta x \text{ in m}.$$

From the regression (with $g = 9.80665\ \mathrm{m\,s^{-2}}$):

$$A = (-0.00106633 \pm 0.000662925)\ \mathrm{m},$$
$$B = (0.406136 \pm 0.0064389)\ \mathrm{m\,kg^{-1}}.$$

Since $B = g/k$,

$$k = \frac{g}{B} = (24.146 \pm 0.38282)\ \mathrm{N\,m^{-1}} \qquad (\text{form } k_{\text{best}} \pm \delta k).$$

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Conclusion

In conclusion, the calculated value of the spring constant of the spring used in this experiment was $(24.146 \pm 0.38282)\ N \cdot m^{-1}$. From a graphical analysis perspective, the fit of the line seems to agree very well with the data points, and the residuals are small compared to the stated uncertainties on extension (all points lie near the weighted least-squares line and within their error bars). The intercept for A being -1.07 mm is reasonable in making the assumption that the spring has no extension with no force applied and the spring is ideal in the linear regime, further strengthening our findings. The reduced χ^2 is well below 1 (0.1), which suggests our extension uncertainties may overestimate the true scatter; the visual agreement of the line with the points is still excellent.